Implementation of IPv6 IOAM

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Netdev 0x14, ~~Vancouver~~ Virtual, ~~March~~ August 2020

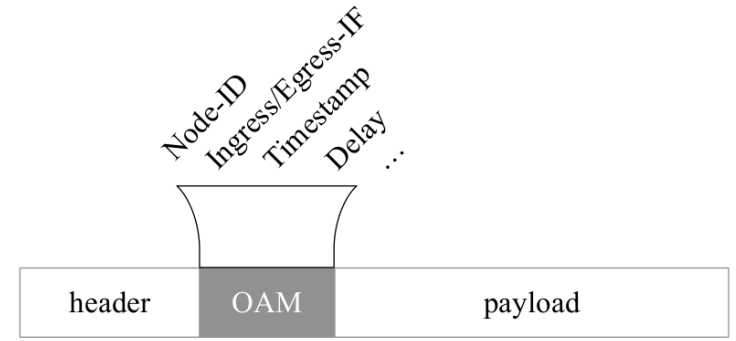
What is IOAM?

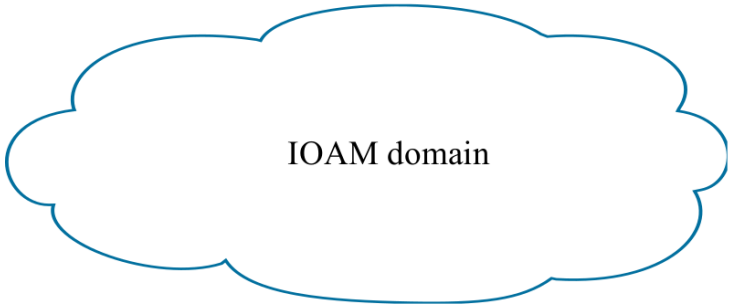
"**I**n-situ" **O**perations, **A**dministration and **M**aintenance

In-situ: data recorded within packets

Hybrid: active and passive OAM type

- IETF draft-ietf-ippm-ioam-data
- IETF draft-ietf-ippm-ioam-flags
- and more…





IPv6 IOAM

IPv6 Extension Headers: Hop-by-Hop and Destination Options

→ **draft-ietf-ippm-ioam-ipv6-options**

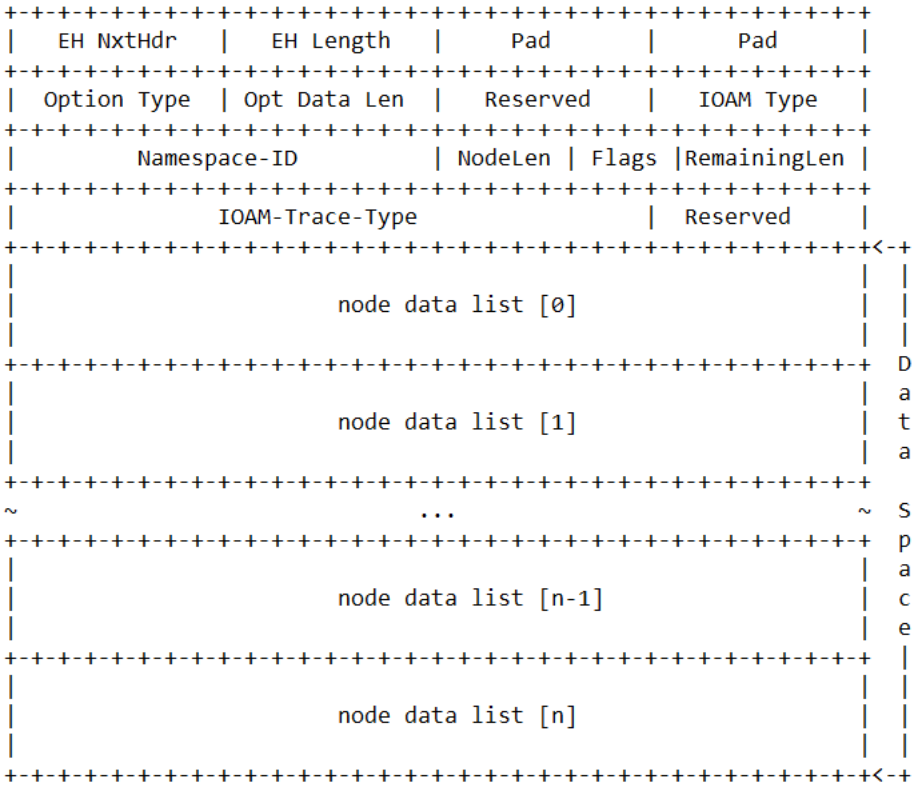
New Hop-by-Hop TLV

New Destination TLV

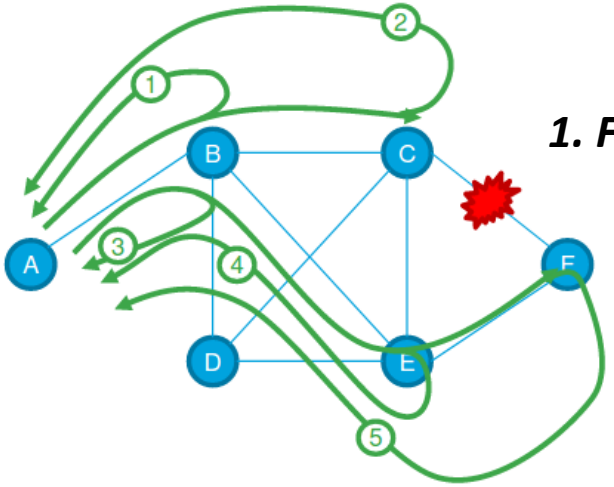
Alignment: **4n**

IOAM option types:

- **Pre-allocated Trace**
- Incremental Trace
- Proof of Transit (POT)
- Edge-to-Edge (E2E)

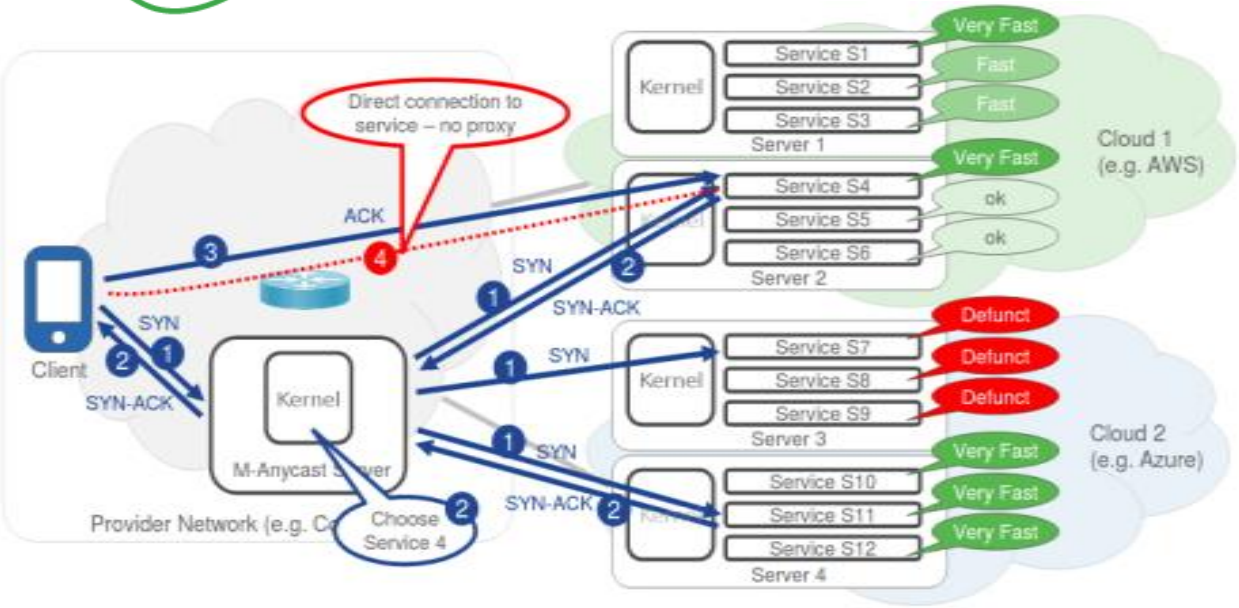


Some interesting use-cases (1)



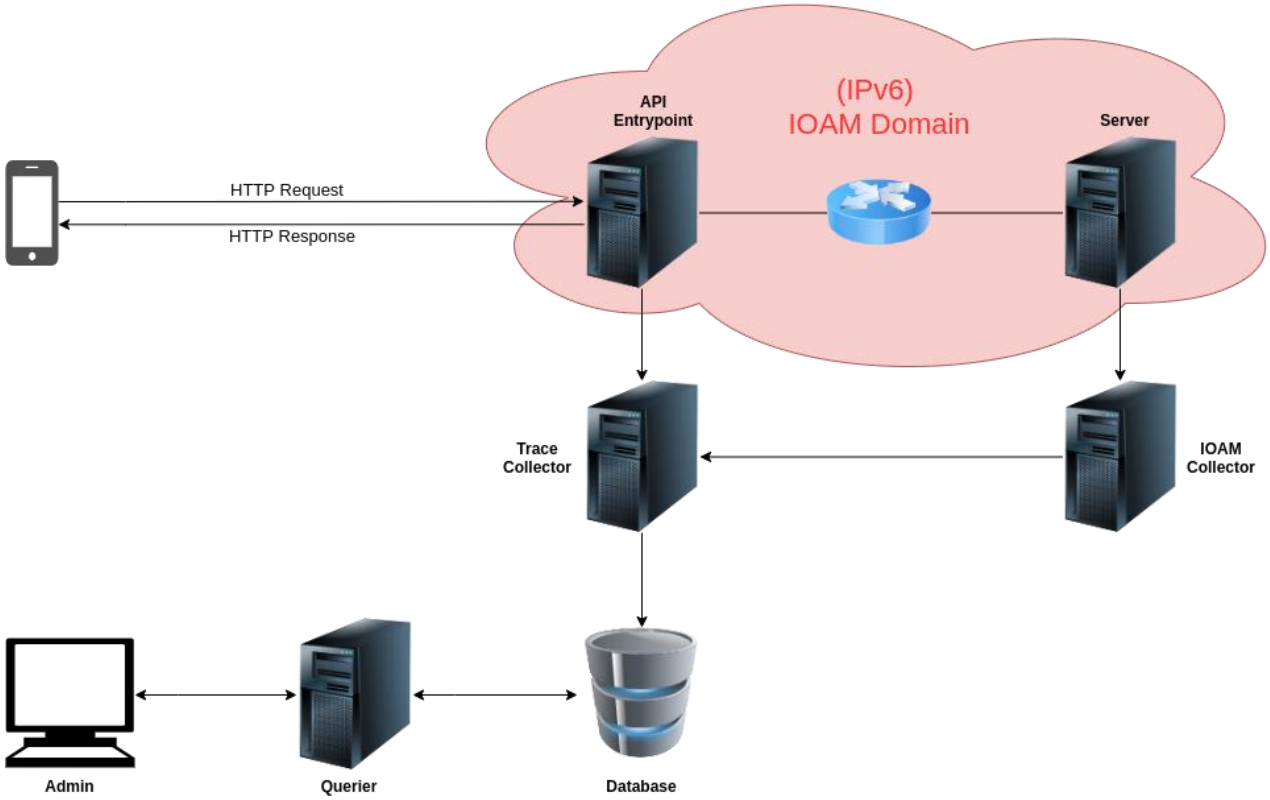
1. Fast failure detection and isolation

2. Smart service selection and load-balancing



Some interesting use-cases (2)

3. Cross Layer Telemetry (CLT) - https://github.com/iurmanj/cross-layer-telemetry



Data plane implementation (1)

https://www.mail-archive.com/netdev@vger.kernel.org/msg333936.html

include/net/netns/ipv6.h

- Hashtable for IOAM namespaces

net/ipv6/exthdrs.c

- Process IOAM Trace options inside a Hop-by-Hop
- Insert IOAM data if IOAM namespace is known
- Remove IOAM Trace option if IOAM namespace must be removed

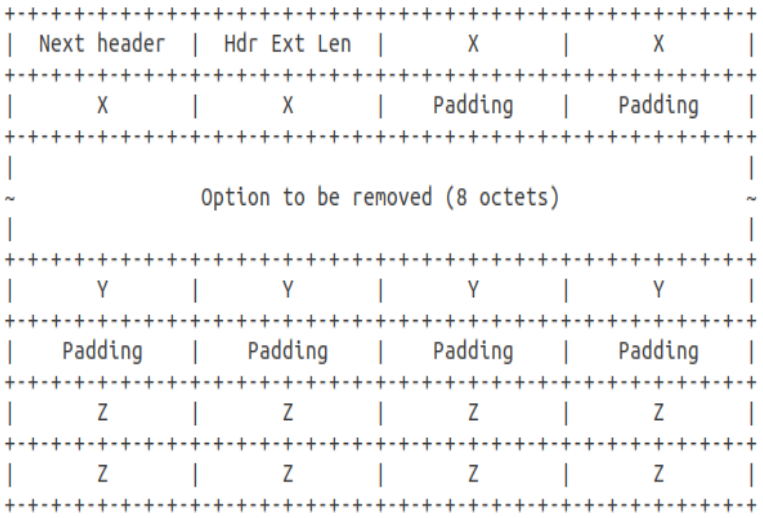
net/ipv6/ip6_input.c

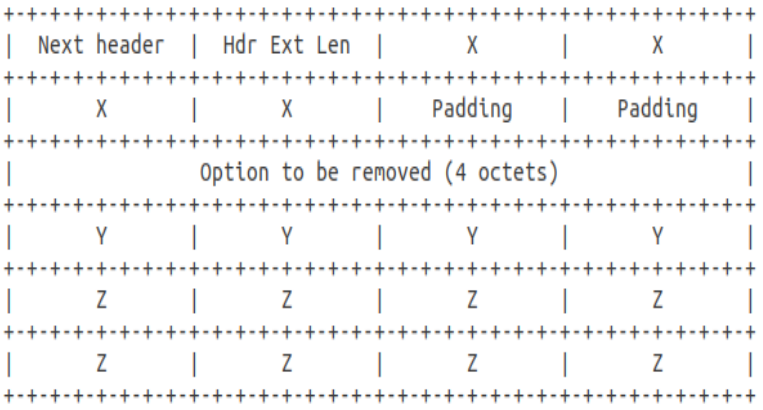
- Remove outer IPv6 header (with IOAM)
- Reinject packet in network stack

Data plane implementation (2)

Remove a TLV and keep correct alignment for every option inside the EH

- *X* option is **2n**-aligned
- *Y* option is **4n**-aligned
- *Z* option is **8n**-aligned





Control plane implementation

RFC 8200 compliant:

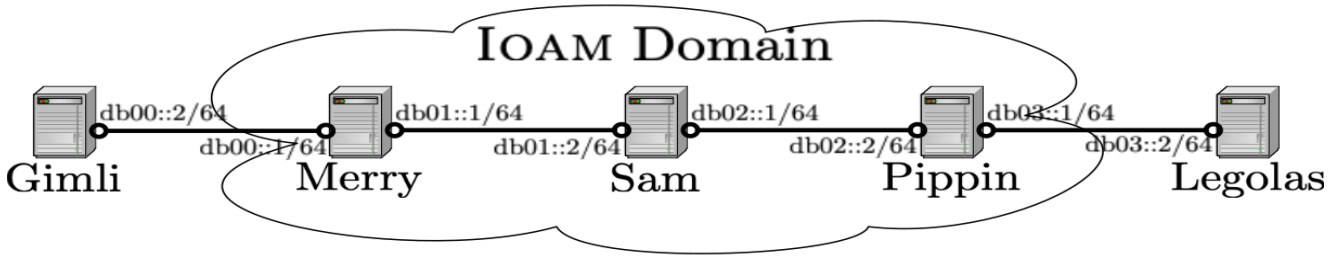
- IOAM **encapsulation** (ingress)
- IOAM **insertion** (end-host)

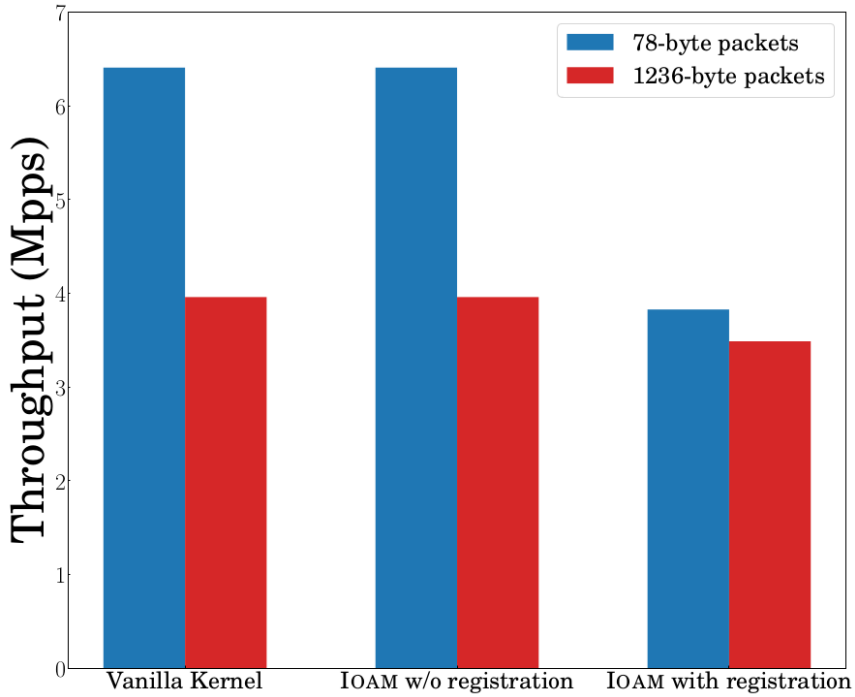


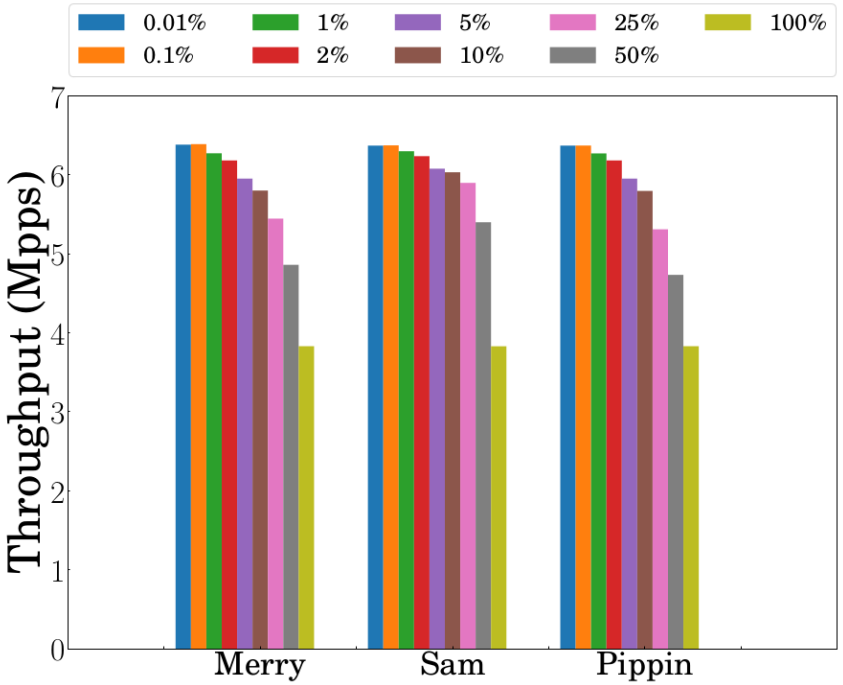
Best solution (?):

1. IOAM buffer attached to a route with Lightweight Tunnels and Route Netlink
2. Generic Netlink:
   a) IOAM buffer stored per network device (new field)
   b) IOAM buffer stored elsewhere

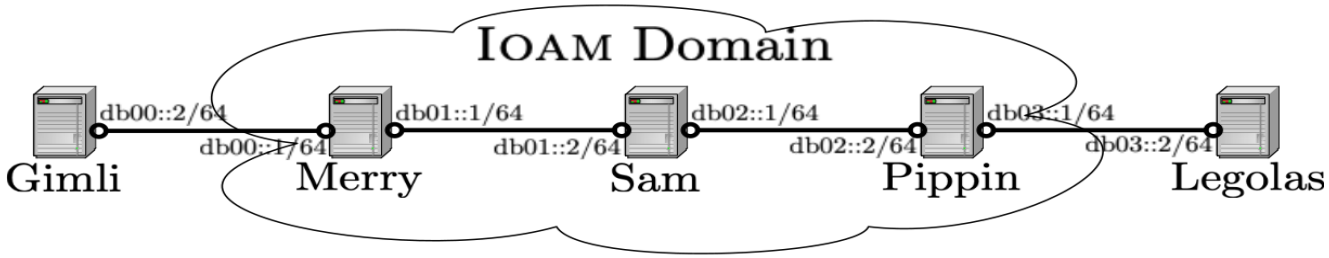
Performances (1)

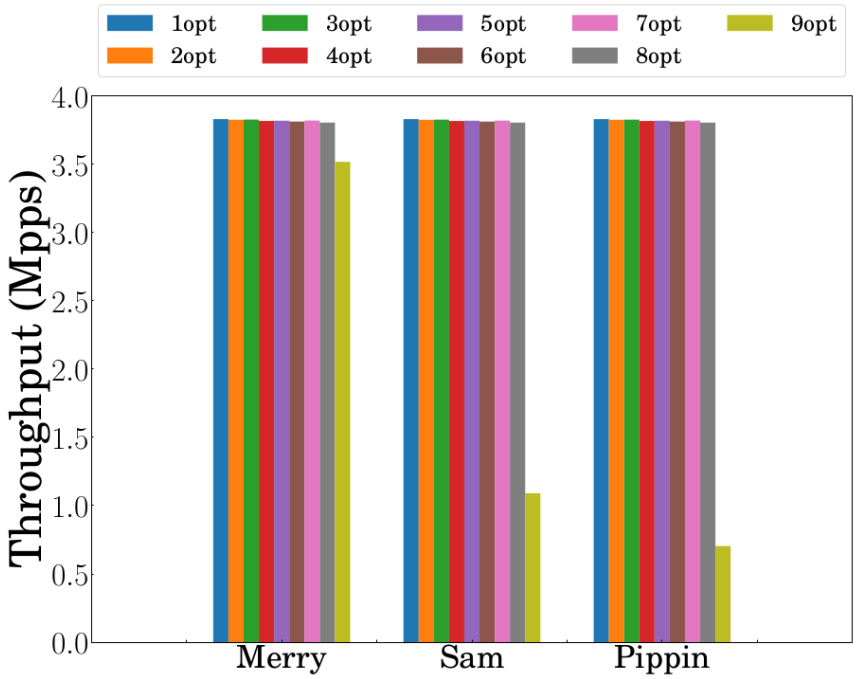


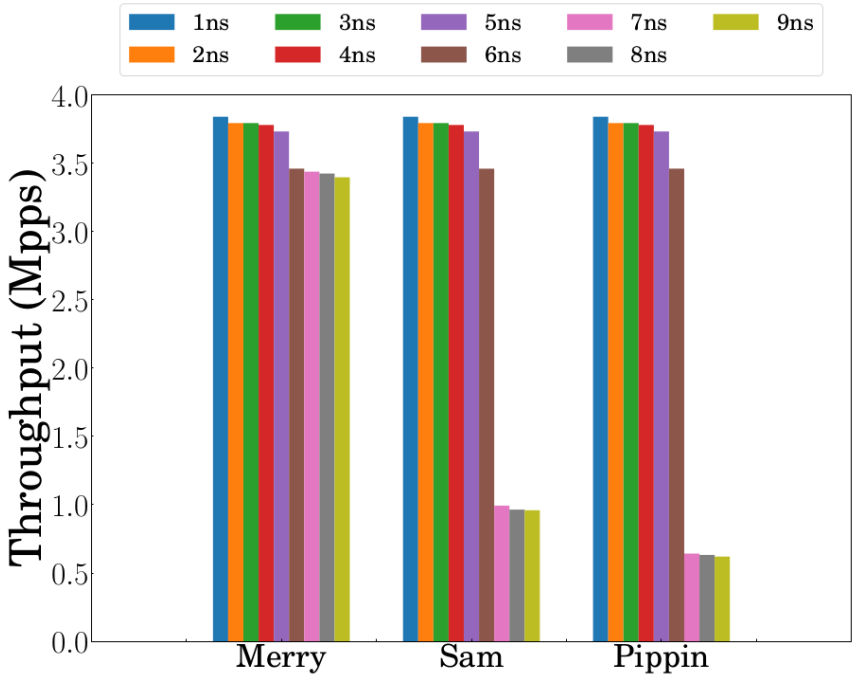




Performances (2)







Conclusion

- Nice feature for the Linux Kernel
- Data plane patch v1 (*net-next*):
  - *https://www.mail-archive.com/netdev@vger.kernel.org/msg333936.html*
- Decision for the control plane
  - What is the best choice?
- Join us during an IETF Hackathon and let’s have fun!

Thank you!

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